A SHEAF REPRESENTABLE OVER AN AFFINE FORMAL ALGEBRAIC SPACE IS A FORMAL ALGEBRAIC SPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The delivered context span is the author's own definition of an affine formal algebraic space, which the statement depends on; it is copied whole.

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Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The presentation Y = colim Y_lambda of an affine formal algebraic space. Same chapter, Definition definition-affine-formal-algebraic-space, invoked at proof lines 4180-4182. Provides the terms Y_lambda and the colimit description of Y.
- A colimit of formal algebraic spaces along the thickening system is a formal algebraic space. Same chapter, Lemma lemma-colimit-is-formal, invoked at proof lines 4185-4186. Upgrades the presentation X = colim X x_Y Y_lambda to the conclusion that X is a formal algebraic space.

Source reference. Original lemma 0AIU: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2973–2973. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original remark 0AIS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2769–2769. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let S be a scheme. We say a sheaf X on $(Sch/S)_{fppf}$ is an *affine formal algebraic space* if there exist

(1) a directed set Λ,
(2) a system $(X_\lambda, f_{\lambda\mu})$ over Λ in $(Sch/S)_{fppf}$ where
   (a) each X_λ is affine,
   (b) each $f_{\lambda\mu} : X_\lambda \to X_\mu$ is a thickening,

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such that

$$X \cong \operatorname{colim}_{\lambda \in \Lambda} X_\lambda$$

as fppf sheaves and X satisfies a set theoretic condition (see Remark 0AIS). A *morphism of affine formal algebraic spaces* over S is a map of sheaves.

Lemma 0.2. *Let S be a scheme. Let Y be an affine formal algebraic space over S. Let $f : X \to Y$ be a map of sheaves on $(Sch/S)_{fppf}$ which is representable by algebraic spaces. Then X is a formal algebraic space.*

Proof. Write $Y = \operatorname{colim} Y_\lambda$ as in Definition 0.1. For each λ the fibre product $X \times_Y Y_\lambda$ is an algebraic space. Hence $X = \operatorname{colim} X \times_Y Y_\lambda$ is a formal algebraic space by Lemma 0AIU. □